JIMMY ZHONG

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EDUCATION

University of Pennsylvania, College of Arts & Sciences
B.A. in Mathematics and Computer Science | GPA: 3.81 / 4.00
Coursework: Algorithms, Data Structures, Discrete Mathematics, Automata & Computability, Machine Learning, Probability

EXPERIENCE

Altverse | Software Engineering Intern — Aug 2026 – Oct 2026
Chrona (chrona.world) — a persistent shared world whose assets, buildings, and scenes are generated by its users from text prompts
- Standardized Chrona's generation pipeline into a serialization layer that lets users chain single assets into persistent, editable scenes.
- Shipped the pipeline into Chrona's live public world and the LA Tech Week showcase, attracting 10 artists with 10K+ followers each.
- Designed and built personalized 3D interactive worlds for artists using Three.js and Tripo assets, acquiring 200K+ views on X.

Welta AI | Founding Full-Stack Engineer — Mar 2026 – Sep 2026
AI advertising campaign management (welta.ai) — 5-engineer founding team, 3 paying business customers, 10+ live campaigns
- Architected the multi-agent campaign manager, cutting cross-platform performance review from 20 manual minutes to a 5-second query.
- Built LLM-driven creative generation flow for programmatic ad creative, accelerating variant generation by 97% (60+ → 2 minutes).
- Built the platform in Next.js and TypeScript with Python LangGraph agents on Supabase Postgres, integrating Meta and Google Ads APIs.

DeepWisdom (MetaGPT) | Software Engineering Intern — Jun 2026 – Aug 2026
Atoms (atoms.dev) — AI website-building agent, 1.2M monthly visits and 70K GitHub stars
- Designed and ran A/B tests routing tailored prompts by user intent archetype, lifting click-through on agent-generated sites by 23%.
- Analyzed a 20,000-user experiment showing style-guidance injection made users 15% more likely to publish, and shipped the feature.
- Built the event-tracking robot for Atoms' experiment, enabling automatic notification of tests and cutting data query time by half.

Franklink | Co-founder & Technical Lead — Sep 2025 – Mar 2026
The first AI networking agent native to iMessage — 2000+ messages, 500+ introductions among users
- Built the agent on LangGraph with a ReAct loop, handled cache in Redis, managed data in Supabase, and deployed the agent on AWS.
- Raised $50K in angel funding from a Y Combinator China founding team member and set the technical direction for a 13-person team.
- Led weekly discussions that broke the roadmap into per-engineer tasks and marketing targets, and reviewed progress with the team.

RESEARCH

Verifier-Guided Self-Training for Large Language Models | PyTorch, Hugging Face, LoRA — Aug 2026 – Oct 2026
Research on how imperfect verifiers shape iterative LLM self-training, submitted to AISTATS 2027
- Proved in a logistic model that two verifiers with identical true- and false-acceptance rates can drive training in opposite directions.
- Engineered a PyTorch and PEFT LoRA pipeline that matches verifiers on acceptance rates, yield, and precision for Qwen3 and Llama-3.2.
- Scaled one-step and multi-round experiments across H100, A100, and RTX GPUs, running 165K greedy evaluations per model.

Migracoast | scikit-learn, GIS, NumPy — Jul 2024 – Sep 2026
Coastal climate-risk modeling across nine U.S. states, released as an open-source research tool
- Built a coastal inundation and climate-risk model for nine U.S. states from sea-level, flood, and housing data, serving 6,500+ users.
- Trained a flood-risk model on 790K FEMA claims with spatially blocked cross-validation, cutting coastal error 27% below a linear model.
- Built the pipeline in Python with scikit-learn, GeoPandas, and netCDF4, serving risk scores through a Flask API at 120 ms per query.

Leakage-Safe Validation Harness for Equity Return Models | scikit-learn, pandas, NumPy — Aug 2026 – Sep 2026
A harness for the leakage bug class that makes models on financial time-series data look strong in development and fail in production
- Built a Python pipeline turning 11 years of daily S&P 500 prices into a cached, 291K-row panel of six standardized factors across 100 names.
- Implemented a purged walk-forward CV for next-month stock return forecasts, identifying 19× overestimation of accuracy by shuffled k-fold.
- Traced the inflation to overlapping 21-day return labels across train and test folds, showing that the effect scales with model capacity.

LEADERSHIP & ACTIVITIES

Wharton China Association, Sponsorship Chair — Lead sponsor outreach for the annual Penn Wharton China Summit. — Aug 2025 – Present
Wharton Hedge Fund Club, Global Macro Board — Research and present macro trade ideas to the coverage group. — Oct 2025 – Present
Penn Venture Lab, Member — Build alongside student founders in Penn's early-stage startup incubator. — Aug 2025 – Present

SKILLS & INTERESTS

Programming Languages: Python, Java, OCaml, C++, TypeScript, SQL, HTML/CSS, LaTeX
AI & ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LoRA/PEFT, RAG, LangGraph, LangChain, multi-agent orchestration
Systems & Tools: Linux, AWS, Redis, PostgreSQL, Docker, Git, Supabase, React, OAuth, pandas, NumPy, Three.js, Blender, QGIS
Natural Languages: English (fluent), Mandarin (native), Japanese (conversational)
Interests: Poker, tennis, squash, guitar, and travel